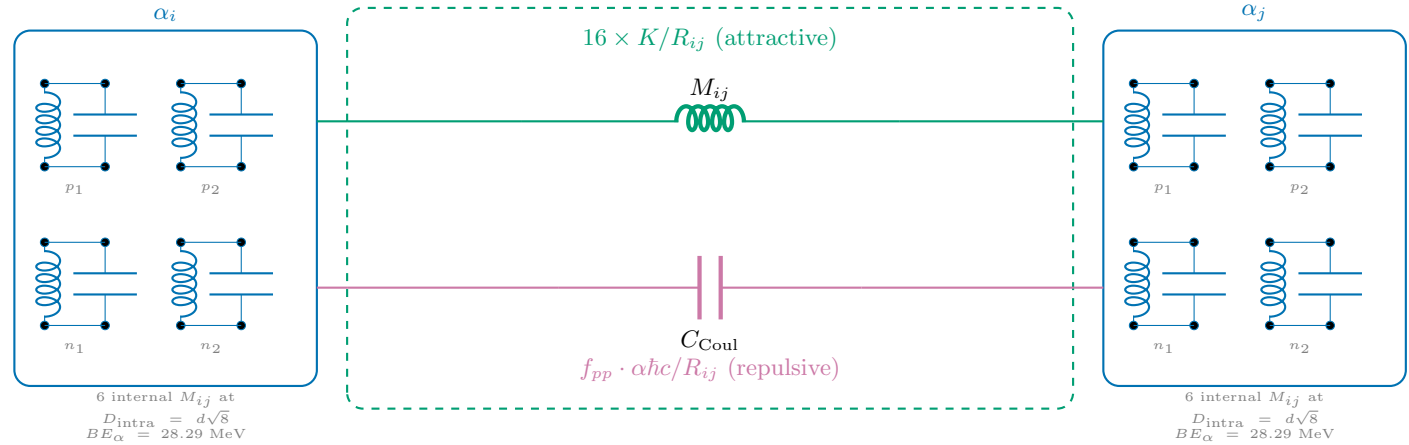


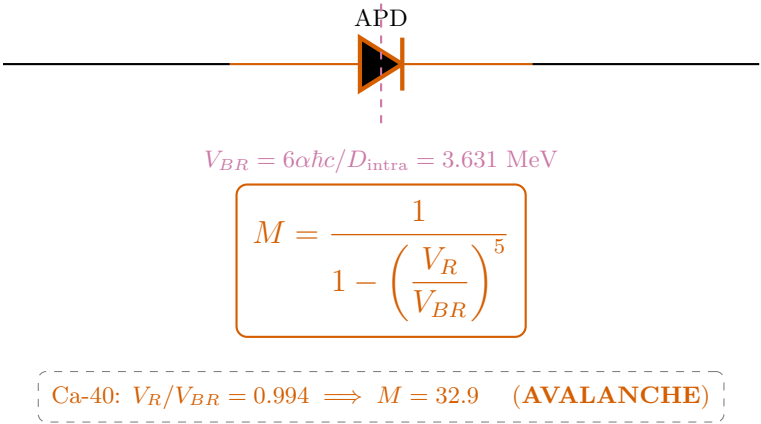
Ca-40 Semiconductor Equivalent Circuit

10 α Bicapped Antiprism — $V_R/V_{BR} = 0.994$ — $M = 32.900$ (Large Signal)

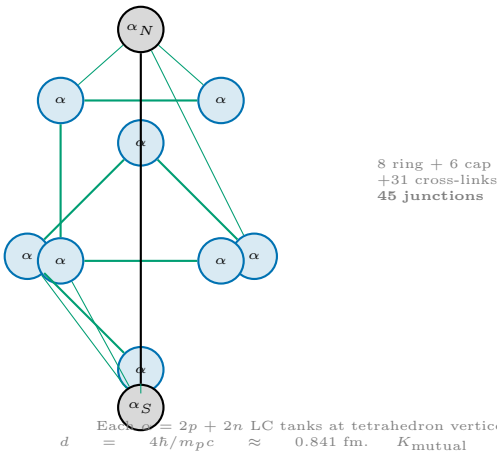
A: Inter-Alpha Junction (Unit Cell)



B: Miller Avalanche Stage



C: Ca-40 Topology (45 junctions)



D: Energy Balance

$$M_{\text{nuc}} = 10 M_\alpha - BE_{\text{net}}$$
$$BE_{\text{net}} = \underbrace{\sum_{i < j}^{45} 16 \cdot \frac{K}{R_{ij}}}_{\text{testStrong(attractive)}} - \underbrace{M \cdot \sum f_{pp} \frac{\alpha \hbar c}{R_{ij}}}_{\text{Coulomb (repulsive)}}$$

Strong: $\sum K/R$ = engine-derived

Coulomb: $\sum \alpha \hbar c / R \times f_{pp}$

Miller: $M = 32.900$ ($V_R/V_{BR} = 0.994$)

Result: 37,322.573 MeV (0.0000% error)

Each $\alpha = 2p + 2n$ LC tanks at tetrahedron vertices ($D_{\text{intra}} = d\sqrt{8} \approx 2.38$ fm).
 $d = 4\hbar/m_p c \approx 0.841$ fm. $K_{\text{mutual}} = 11.34$ MeV·fm. $V_{BR} = 3.631$ MeV.